DeepSeek-V2: A Strong, Economical, and Efficient MoE Language Model

236B Total Parameters • 21B Activated per Token • 128K Context Length

arXiv:2405.04434

The LLM Scaling Dilemma: Performance vs. Training Cost and Inference Speed

Training Cost

Larger dense models incur prohibitive pretraining expenses. Scaling beyond 70B parameters demands massive compute budgets that most organizations cannot afford.

KV Cache Bottleneck

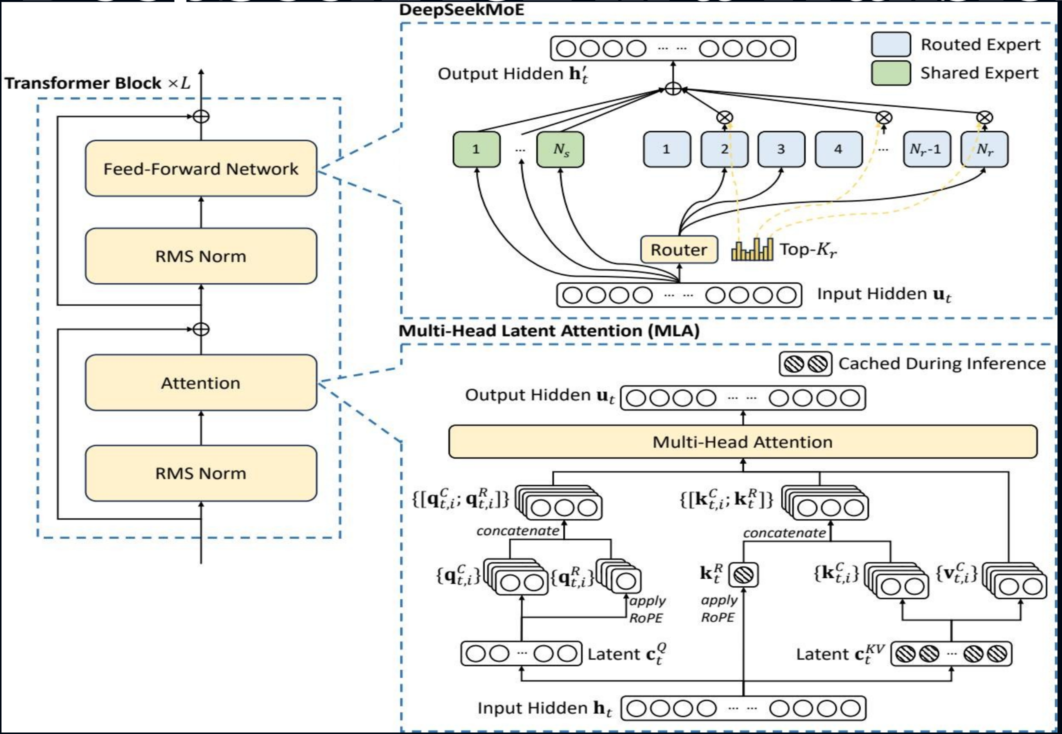
Standard Multi-Head Attention stores full key-value pairs per head per token, creating memory pressure that severely limits batch sizes and context lengths during autoregressive generation.

Inference Speed

Dense activation of all parameters for every token yields low throughput. The gap between model capability and serving efficiency widens as models scale.

Core Question: How can we achieve top-tier performance without paying the full cost of dense scaling?

DeepSeek-V2 Architecture: MLA + DeepSeekMoE in a Transformer Block



Multi-Head Latent Attention (MLA)

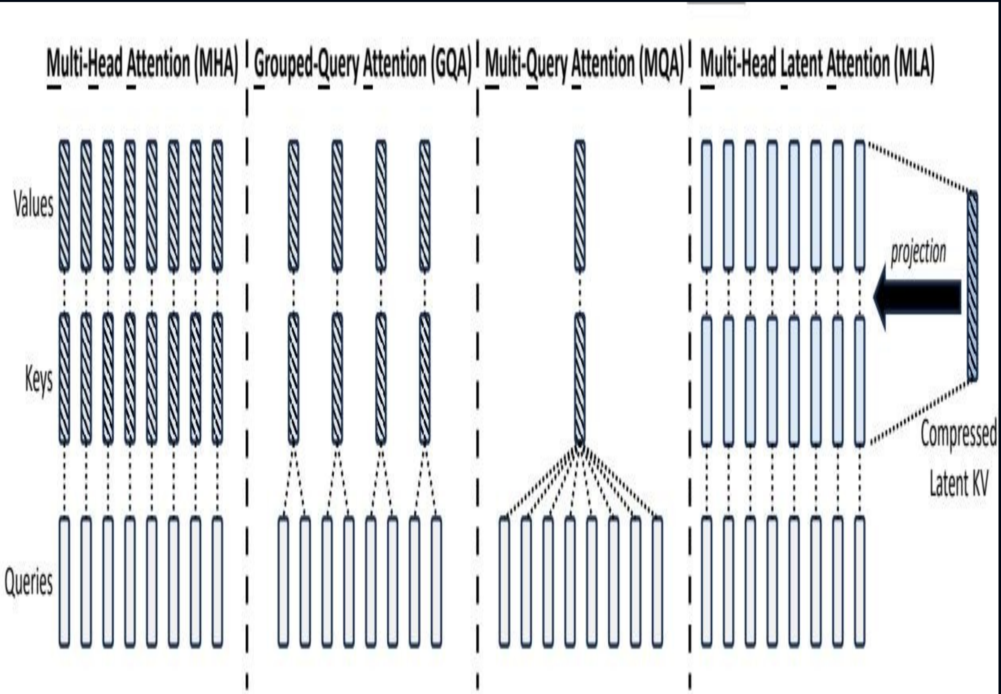
Replaces standard MHA with low-rank KV joint compression. Caches a compact latent vector instead of full K/V tensors, then reconstructs on-the-fly via learned projections.

DeepSeekMoE FFN

Shared experts capture common knowledge; routed experts are sparsely activated via top-K routing. Device-limited routing reduces cross-node communication.

Total: 236B params | Activated: 21B per token | Context: 128K tokens

Multi-Head Latent Attention Compresses KV Cache by 93.3%



How MLA Works

- **Low-rank KV joint compression**

Instead of caching full K and V vectors per head, MLA compresses them into a compact latent vector cKV.

- **On-the-fly reconstruction**

Full keys, values, and a decoupled RoPE-based positional key are reconstructed via learned projection matrices during inference.

- **Better than MHA, less cache than GQA/MQA**

MLA achieves superior performance to standard MHA while requiring dramatically less KV cache than GQA or MQA.

KV Cache Reduction: 93.3% — Performance: better than MHA

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

Expert Design

- **Shared Experts (Ns)**

Always activated; capture common knowledge across all tokens, reducing redundancy in routed experts.

- **Routed Experts (Nr)**

Sparsely activated via a top-Kr router. Only a subset is selected per token, keeping compute low despite 236B total parameters.

- **Fine-Grained Segmentation**

Smaller, more numerous experts increase specialization potential and routing flexibility.

Routing & Load Balancing

- **Device-Limited Routing**

Each token is constrained to experts on at most M devices, minimizing cross-node communication overhead in distributed training.

- **Auxiliary Losses**

Load-balance losses enforce even distribution across experts and devices, preventing hotspot collapse.

- **Token Dropping**

A training-only strategy drops overflow tokens from overloaded experts without affecting inference behavior.

Result: 236B-parameter model trained at 42.5% lower cost than a 67B dense model

42.5% Training Cost Reduction and 5.76× Generation Throughput vs. DeepSeek 67B

DeepSeek 67B (Dense Baseline)

- 67B activated parameters per token
- Standard Multi-Head Attention
- Dense FFN layers (no sparsity)
- Higher training cost baseline
- Larger KV cache footprint
- Lower generation throughput

DeepSeek-V2 (MoE + MLA)

42.5%	Training cost reduction
93.3%	KV cache reduction
5.76×	Higher max generation throughput

Relative improvements vs. DeepSeek 67B dense predecessor. All metrics from arXiv:2405.04434.

DeepSeek-V2 Tops MMLU Among Open-Source Models at Only 21B Activated Parameters

MMLU Rankings

Model	Act. Params	MMLU
DeepSeek-V2	21B	78.5
LLaMA 3 70B	70B	77.0
Mixtral 8×22B	39B	77.6
Qwen1.5 72B	72B	75.6
DeepSeek 67B	67B	73.0
LLaMA 2 70B	70B	69.0
Mixtral 8×7B	13B	70.6
LLaMA 3 8B	8B	65.0

Pareto-optimal: highest MMLU at lowest activated parameter count

DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Benchmarks

Alignment Pipeline

SFT on 1.5M conversational sessions → Reinforcement Learning via Group Relative Policy Optimization (GRPO)

AlpacaEval 2.0

Length-Controlled Win Rate

38.9%

MT-Bench

Overall Score

8.97

AlignBench (Chinese)

Overall Score

7.91

Strongest open-source chat model in Chinese | Competitive with closed-source systems in English

Key Takeaways: Architectural Innovation Unlocks Efficient Scaling

1. MLA is a strictly better KV-cache solution: superior to MHA in quality, yet cuts KV cache by 93.3% — making GQA and MQA obsolete for inference efficiency.

2. DeepSeekMoE enables economical training of massive models: 236B parameters trained at 42.5% lower cost than a 67B dense model via shared experts and fine-grained routing.

3. Top-tier performance with minimal activation: only 21B activated parameters match or exceed 70B-class dense models on MMLU — a Pareto-optimal trade-off.

4. 5.76× throughput improvement over DeepSeek 67B makes large-scale serving practical, closing the gap between capability and deployment efficiency.

5. Strong multilingual alignment: DeepSeek-V2 Chat (RL) is the strongest open-source chat model in Chinese and competitive with closed-source systems in English